

Deep Learning Fundamentals

Neural Networks

A neural network is composed of layers of interconnected nodes called neurons. Each connection has a weight that is adjusted during training. Deep learning refers to networks with many hidden layers that can learn hierarchical representations of data.

Gradient Descent

Gradient descent is a first-order optimization algorithm. It minimizes a loss function by iteratively moving in the direction of the steepest descent, which is the negative gradient. The learning rate controls the size of each step taken during optimization.

Backpropagation

Backpropagation computes the gradient of the loss with respect to each weight by applying the chain rule from the output layer back to the input layer. This enables efficient training of deep networks.